

Phase G Decision Memo

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1. The objection that is still open

Every other major reviewer criticism has been answered. This one has not.

Qwen3-30B-A3B withholds domain-specific harmful content on 65.87% of English prompts but only 22.82% of matched Romanized-Hindi prompts – a **43.06 pp** non-assistance gap (95% CI 38.49-47.62). Equivalence certification controls *what* is asked: intent, scenario, information level, attack strategy, cultural specificity, target group, ambiguity, strategy structure.

It does not control **how far the surface form sits from the model’s training distribution**.

So a reviewer can still say: *this is not about Hinglish, it is about unusual input, and Hinglish happens to be unusual*. Nothing in the frozen design can answer that, because every existing arm is either ordinary English or Romanized Hindi. It is the objection behind Reviewer 2’s score of 2.

2. What Phase G tests

A third arm on the same frozen base pairs:

- **E** ordinary English (already collected)
- **R** Romanized Hindi / Hinglish (already collected)
- **U** semantically matched **unusual English** (new)

U is unambiguously English. If the effect is about the Hinglish register, U behaves like E. If it is about surface-form novelty, U behaves like R.

The primary estimand is a paired difference-in-differences, per model:

$\Delta = (E - R) - (E - U) = \text{mean}(1[s(U)=0] - 1[s(R)=0])$
 $\rho = (E - U) / (E - R)$ the share of the RH effect that unusual English reproduces

Because the English arm is shared it cancels, so the primary test is an exact paired test on U-versus-R discordant pairs. Clean, well powered, distribution-free.

3. Recommended design

N	96 base pairs , 8 in each of the 12 category x strategy cells
Unusual English	two balanced sub-registers: archaic/over-formal, and telegraphic/clipped
Selection	SHA-256 rank of pair identity, frozen manifest, outcome-independent
Certification	DeepSeek V4 Flash + GPT-5 Mini, dual hard gate, English-to-English axes
Target inference	U arm only, same three models, temperature 0, 4,096 tokens, empty system prompt
Judging	Gemini 2.5 Flash on U; E and R judgments reused unchanged
Primary model	Qwen only – the design is not powered for the other two, and says so

Why 96 and not 72 or 120. The discriminating scenario is U reproducing three quarters of the RH effect, which is the hardest case to distinguish from generic OOD. Power there is 0.748 at N=72 (a reviewer would rightly call that underpowered), **0.913 at N=96**, and 0.971 at N=120. N=120 buys 0.058 more power for

25% more inference. The three selections are nested, so 96 can be extended to 120 later without invalidating anything.

4. What already exists and is reused

- All 504 frozen English and Romanized-Hindi target responses and their Gemini judgments. The 96 selected pairs contribute 192 existing responses at no cost.
- The generator, auditor and judge harnesses, prompts, parsers and rubric.
- The certification contract, adapted to English-to-English.
- The pair-clustered statistical machinery from Phases B-D.

Only the U arm needs new inference.

5. New inference and cost

	N=72	N=96	N=120
New target calls	228	304	380
Total new API calls (all stages)	~650	~ 864	~1,080
Expected cost	\$1.36	\$1.82	\$2.27
Expected-high cost	\$2.54	\$3.39	\$4.24
Hard ceiling	–	\$10.00	–

Every generation, certification and judging unit cost is *measured* from this project’s own billing ledgers, not from list prices. Expected-high assumes the V2 bank-build funnel reproduces unchanged **and** an extra GPT-5 Mini cross-judge is added. Derivation: `analysis/phase_g/COST_MODEL.md`.

Dominant cost: GPT-5 Mini secondary certification, at \$0.005932 per call – eight times the DeepSeek rate, and \$7.07 of the \$11.84 that built the entire V2 bank. Dropping to DeepSeek-only certification would save about \$0.75. **We recommend against it.** The recovered V1 evidence shows single-auditor certification is exactly the failure that made V1 unusable; reintroducing it to save under a dollar would be indefensible.

For scale: Phase D cost \$3.68. Phase G at N=96 costs about half that.

6. Possible outcomes and what each would mean

The interpretation rule is fixed in advance, on the 95% interval for **rho**:

Result	Reading	Consequence for the paper
CI upper bound < 0.50	Register-specific dominant. Unusual English does not reproduce most of the RH effect.	The strongest outcome. The language-conditioned framing is vindicated against the main standing objection, and the paper can say so.
CI lower bound > 0.50	Generic-OOD dominant. Unusual English reproduces most of it.	The paper must be reframed around distributional novelty rather than Hinglish. Painful, but it is the truth and far better found by us than by a reviewer.
CI spans 0.50, excludes 0 and 1	Mixed. Both mechanisms contribute.	Report both components; claim neither exclusively. Still a substantially stronger paper than silence.
CI spans both 0 and 1	Inconclusive.	Report the interval and claim nothing. Do not re-run at a different N to chase a threshold.

We commit in advance to publishing whichever occurs, including the one that weakens the headline.

7. Limitations Phase G will not remove

- **Not powered for GPT-OSS or Nemotron.** Their E-R gaps (4.56 and -1.79 pp) are too small; power never exceeds 0.21 at any N considered. Stated up front so their nulls cannot later be dressed up as findings.
- **Two sub-registers, not all of English.** It shows Hinglish differs from these two English sub-registers, not that it is unique among all registers.
- **No mechanism.** It is a behavioural contrast, not an account of what happens inside the model.

- **A third shared property** common to both R and U could still drive the effect.
- **Same rubric, same judge family.** Phase G inherits the measurement limitations of the frozen design, with one improvement: E and U are both judged under the **en** language tag, so the E-U contrast is made *within* one judge condition. The E-R contrast never was. That is a genuine partial repair of the judge-blinding limitation.

8. Relationship to Phase E

Phase G does **not** depend on Phase E succeeding, and can be run and analysed whatever the human labels show. Three interpretation points should be revisited once labels arrive:

1. Poor human-Gemini agreement on Qwen would mean stating Phase G conclusions in terms of *judge-measured* non-assistance.
2. A systematic human-judge offset largely cancels in a difference of differences; a *condition-dependent* offset would not, and would warrant a small human check on U responses.
3. If Phase E proves unusable, Phase G still runs; both are then reported as open.

Nothing in Phase G's selection, generation, certification, inference or primary analysis may change in response to Phase E.

9. Recommendation

Fund it. Run N=96.

Three reasons.

It is the last substantive scientific objection. Everything else the reviewers raised has been answered by the reconstruction, the audit and the verification gate. This one is untouched, and it is the one that a competent reviewer will raise first.

It costs about two dollars. Expected \$1.82, expected-high \$3.39, hard ceiling \$10.00 with an automatic abort. This is not a resource-allocation question; it is a question of whether we want the answer.

The downside risk is the point, not a reason to avoid it. If unusual English reproduces most of the effect, the paper's framing is wrong, and we would much rather discover that ourselves than have it discovered in review. A preregistered control that we commit to reporting either way is exactly what makes the register-specific claim credible if it survives.

The one thing not to do is submit with this open and hope no reviewer presses it.

Awaiting explicit authorization. Nothing will be executed until it is given.